Computer Science & DA

Warehousing

Lecture -1

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Topics to be Covered

- Topic: Introduction
- Topic: Steps in Datawarehouse
- Topic: ETL
- Topic: ELT Based Warehousing

GATE – 2009 3rd year 682 AIR

4th year 19 AIR

IISc B'lor

since 2011 GATE Teaching

since 2007 Engg. Teaching

MTech from BITS Pilani (D.S.)

→ Teaching

→ Fitness Trainer

Topic : Course Structure

Topics
• Basics
• Datawarehouse Schema
• Data Transformation
• Data Discretization
• Data Sampling
• Data Compression
• Datawarehouse Measures

Topic : Data

raw facts & figures

Topic : Database

The collection of data, usually referred to as the database, contains information relevant to an enterprise.

example of college Database:-

students

Rno.	name	father name	dob....
1	—	—	.....
2	—	—	—
⋮			

fees

Rno	paid amount	date	mode of payment
1	10000	—	—
2			
3			

Book_issued

Rno.	Bid	issued_date

Books

Bid	Book_title	Author	@

Topic : DBMS → s/w tool

A database-management system (DBMS) is a collection of interrelated data and a set of programs to access those data.

Topic : Data Warehousing

In computing, a data warehouse, also known as an enterprise data warehouse, is a system used for reporting and data analysis and is considered a core component of business intelligence.

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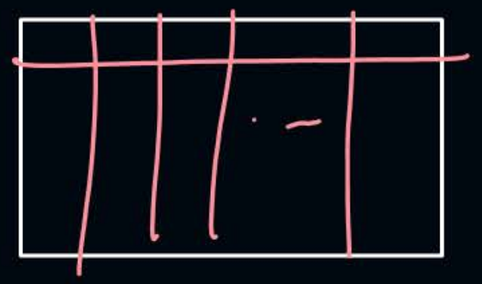
Data warehouses are central repositories of integrated data from one or more different sources. They store current and historical data in one single place.

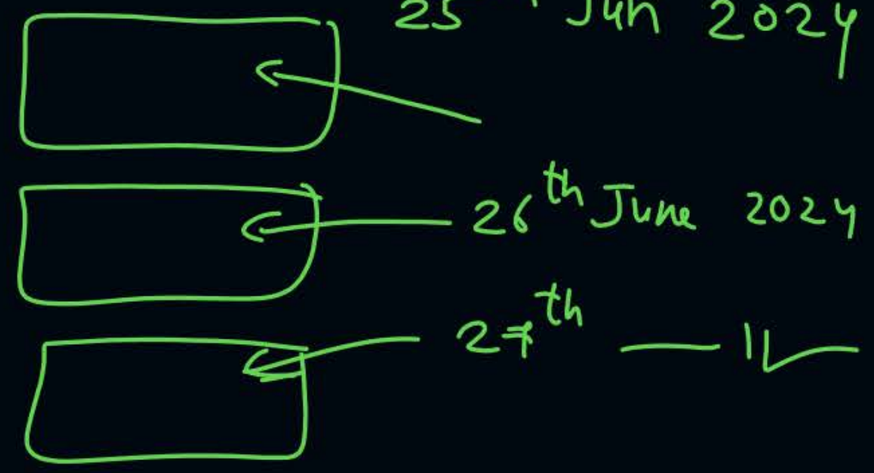
Database

Reliance digital



Transactional data

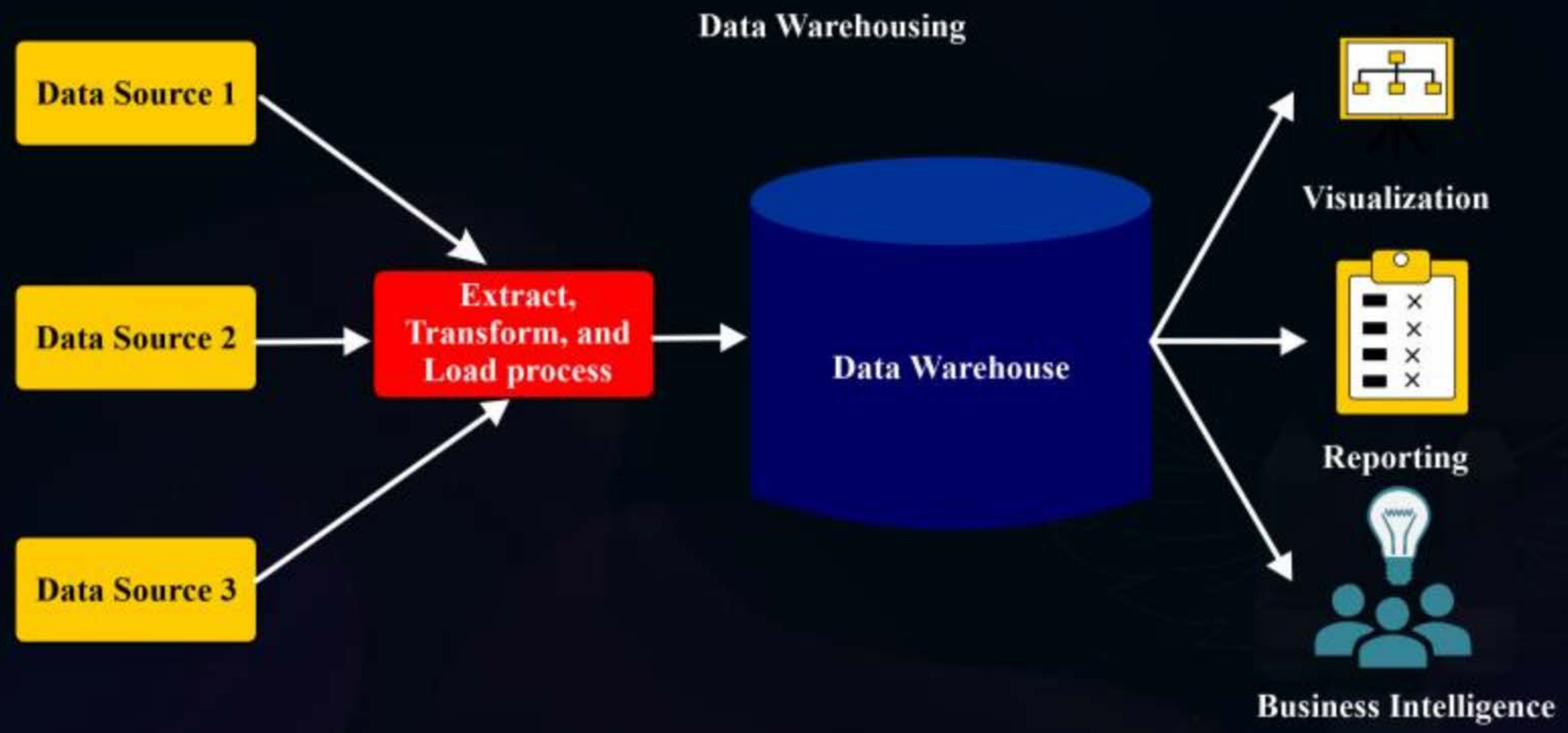
Datawarehouse



Historical data

No Transactions here

Topic : Data Warehousing



Topic : Steps in Data Warehousing

1. Extraction of data
2. Cleaning of data
3. Conversion of data
4. Storing in a warehouse

Topic : ETL (Extract, Transform, Load)

Three-phase process where data is extracted, transformed and loaded into an output data container.

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Three-phase process where data is extracted, transformed and loaded into an output data container.

Done by application software but can be done manually also

Topic : ETL Tools

- Integrate.io
- IBM DataStage
- Oracle Data Integrator
- Fivetran
- Coupler.io
- SAS Data Management
- Talend Open Studio
- Whatagraph
- Pentaho Data Integration
- Singer
- Hadoop
- Dataddo
- AWS Glue
- Azure Data Factory
- Google Cloud Dataflow
- Stitch
- Informatica PowerCenter
- Skyvia
- Portable

2 mins Summary

Topic — Introduction

Topic — Steps in Datawarehouse

Topic — ETL Based Warehousing

Topic — ELT Based Warehousing

Happy Learning

THANK - YOU